# Exit Ticket · reflect and retrieve

## Exit Ticket

Big O directly tells you the elapsed seconds of a program.

## Use key vocabulary from memory

1. Explain O(1) in your own words.

2. Explain O(n) in your own words.

3. Explain O(log n) in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 074 · Big O: Constant, Linear and Logarithmic

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is an algorithm's input size n?

### Q2

What is a trace table?

### Q3

What is an invariant?

### Q4

A sorted search set of 128 candidates is halved repeatedly to one. How many halvings? Show the sequence.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Observe — Big O describes the upper bound on how work grows with input size, not the precise milliseconds of an execution.

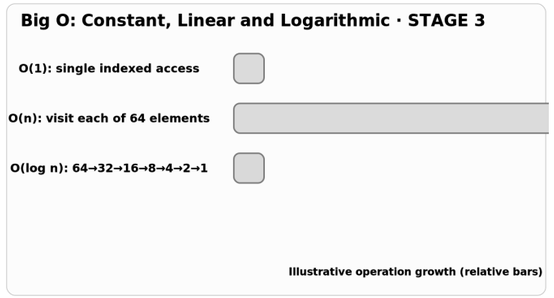
## Core idea 2

Explain — A fixed direct index operation is O(1), scanning n values is O(n), and repeatedly halving n is O(log n).

## Core idea 3

Evaluate — For n=64, a linear scan may inspect 64 items, while halving 64→1 takes six halvings. Sorting requirements and constants still matter.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain why O(log n) binary search may be unsuitable for unsorted data even if asymptotically faster than linear search. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Match direct array access, full array scan and repeated halving with Big O. [6]

## Question 2

For input size64, compare scan steps64 and repeated halving to1; calculate halvings. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

O(1) — Constant asymptotic work.

O(n) — Linear growth.

O(log n) — Logarithmic growth.

asymptotic — Behaviour as size becomes large.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

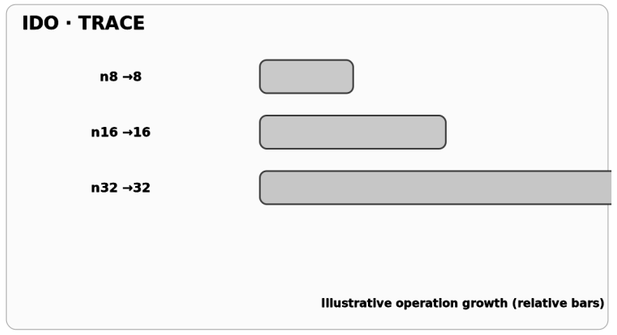
## Problem

Classify loop performing one action for each item in n-element array.

## Follow the worked method

1. Explain — The number of operations doubles when n doubles.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Explain
Each iteration cuts remaining size in half.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.